Rockchip DP Signal Test Guide

ID: RK-KF-YF-C13

Release Version: V1.0.0

Release Date: 2026-06-08

Security Level: □Top-Secret □Secret □Internal ■Public

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Preface

Overview

This document provides Register configuration guidance for DP PHY signal testing on RK3576 and RK3588.

Product Version

Chip Models	Kernel Version
RK3576/RK3588	Linux-5.10/Linux-6.1

Target Audience

This document (this guide) is primarily intended for the following engineers:

Technical Support Engineers

Software Development Engineer

Revision History

Version	Author	Date	Modification Description
V1.0.0	Zhang Yubing	2026/6/8	Initial version

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1. DP PHY Overview

The RK3576 and RK3588 DP PHYs are shared PHYs for DP and USB (USBDP PHY). This document describes the signal metric adjustment methods when using the USBDP PHY for DP functionality.

The base addresses for the RK3576 and RK3588 USBDP PHYs are as follows:

USBDP PHY	USBDP PHY Base Address
RK3588 USBDP PHY0	0xfed88000
RK3588 USBDP PHY1	0xfed98000
RK3576 USBDP PHY	0x2b018000

When configuring Registers, the Register address must be calculated by adding the Register offset to the corresponding PHY base address.

2. DP Signal Tuning

- **The IO commands in this document use RK3588 PHY0 Lane0 as an example. Tuning other Lanes requires modifying the Register offsets; tuning RK3588 PHY1 and RK3576 PHY requires modifying the base address.**
- **A single PHY Register is typically used for multiple function configurations. If multiple Tuning items require joint adjustment, the Register value must first be read, then ORed with the bits for the new Tuning settings before being written back. Otherwise, previously tuned metrics will be overwritten.**

2.1 DP Amplitude Control

2.1.1 DP TX Driver Main-Tap Level

Adjust voltage; it is recommended to prioritize adjusting this Register.

- Register offset

```
// Bit 5 is the enable bit, default={0x0f}
lane0: 0x0810[5:0]
lane1: 0x1010[5:0]
lane2: 0x1810[5:0]
lane3: 0x2010[5:0]
```

- Adjustment and Range

The following IO commands are for Lane0. For other Lanes, please modify the register offset; for example, the address for Lane1 is 0xfed89010 (0xfed88810 + 0x800).

Amplitude	**Command [reg offset 0x0810, bit 5:0]**
4'b1010: 1200mv (max)	io -4 0xfed88810 0x2a
4'b1001:	io -4 0xfed88810 0x29
4'b1000:	io -4 0xfed88810 0x28
4'b0111:	io -4 0xfed88810 0x27
4'b0110:	io -4 0xfed88810 0x26
4'b0101:	io -4 0xfed88810 0x25
4'b0100:	io -4 0xfed88810 0x24
4'b0011:	io -4 0xfed88810 0x23
4'b0010:	io -4 0xfed88810 0x22
4'b0001:	io -4 0xfed88810 0x21
4'b0000: 400mv (min)	io -4 0xfed88810 0x20

2.1.2 DP TX PMOS Current Control

Adjust the current. It is recommended to configure this register only when voltage adjustment fails to meet requirements.

- Register Offset

```
// Enable bit[1:0], default={0x1, 0xe7}
lane0: Enable: 0x0818[1:0]; Adjustment: 0x081c[7:5]={3'b000 ~ 3'b111}
lane1: Enable: 0x1018[1:0]; Adjustment: 0x101c[7:5]
lane2: Enable: 0x1818[1:0]; Adjustment: 0x181c[7:5]
lane3: Enable: 0x2018[1:0]; Adjustment: 0x201c[7:5]
```

- Adjustment and Range

The following IO commands are for Lane0. For other Lanes, please modify the register offset; for example, the address for Lane1 is 0xfed89010.

Amplitude	Command [reg offset 0x081c, bit 7:5]
3'b111:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0xe5
3'b110:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0xc5
3'b101:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0xa5
3'b100:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0x85
3'b011:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0x65
3'b010:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0x45
4'b001:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0x25
4'b000:	io -4 0xfed88818 0x03 io -4 0xfed8881c 0x05

2.2 DP Equalization

2.2.1 DP TX De-emphasis

- Register Offset

```
// bit4 is the enable bit, default=0x0, i.e., 0dB
lane0: 0x0814[4:0]
lane1: 0x1014[4:0]
lane2: 0x1814[4:0]
lane3: 0x2014[4:0]
```

- Adjustment and Range

The following IO commands are for Lane0. For other Lanes, please modify the register offset; for example, the address for Lane1 is 0xfed89010 (0xfed88814 + 0x800).

Level	Command [reg offset 0x0814, bit 4:0]
5'b10000: 0 dB	io -4 0xfed88814 0x10
5'b11011: -5 dB (Recommend to use over 5 Gbps speed)	io -4 0xfed88814 0x1b
5'b10101: -2.8 dB (Recommend to use over 10 Gbps speed)	io -4 0xfed88814 0x15
5'b11001: -6.9 dB (not recommended)	io -4 0xfed88814 0x19

2.2.2 DP TX Preshoot Level

- Register Offset

```
// Bit 6 is the enable bit, default=0x0
lane0: 0x0818[6:2]
lane1: 0x1018[6:2]
lane2: 0x1818[6:2]
lane3: 0x2018[6:2]
```

- Adjustment and Range

Note: This register is multiplexed with TX PMOS current control.

The following IO commands apply to Lane0. For other lanes, modify the register offset accordingly; for example, the address for Lane1 is 0xfed89018 (0xfed88818 + 0x800).

Level	Command [reg offset 0x0818, bit 6:2]
5'b1000000: 0 dB (default, recommend for speeds over 5 Gbps)	io -4 0xfed88818 0x40
5'b1011000: -1.6 dB (recommend for speeds over 10 Gbps)	io -4 0xfed88818 0x58
5'b11001: -6.9 dB (not recommended)	io -4 0xfed88818 0x64

2.3 DP TX Slew Rate

2.3.1 DP Faster Slew Rate Control

- Register Offset

```
// default={0xe7, 0x60}
lane0: Enable: 0x081c[1:0]={2'b11}; Drive Strength: 0x0820[5:3]={3'b111 ~ 3'b000}
lane1: Enable: 0x101c[1:0]; Drive Strength: 0x1020[5:3]
lane2: Enable: 0x181c[1:0]; Drive Strength: 0x1820[5:3]
lane3: Enable: 0x201c[1:0]; Drive Strength: 0x2020[5:3]
```

- Adjustment and Range

The following io commands are for Lane0; for other Lanes, please modify the register offset. For example, the address for Lane1 is 0xfed8901c (0xfed8881c + 0x800).

Amplitude	**Command [reg offset 0x081c, bit 7:5]**
3'b111:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x78
3'b110:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x70
3'b101:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x68
3'b100:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x60
3'b011:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x58
3'b010:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x50
4'b001:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x48
4'b000:	io -4 0xfed8881c 0xe7 io -4 0xfed88820 0x40

2.3.2 DP Slower Slew Rate Control

- Register Offset

```
// default={0xe7, 0x60}
lane0: Enable: 0x081c[1:0]={2'b10}; Drive Strength: 0x0820[5:3]={3'b111 ~ 3'b000}
lane1: Enable: 0x101c[1:0]; Drive Strength: 0x1020[5:3]
lane2: Enable: 0x181c[1:0]; Drive Strength: 0x1820[5:3]
lane3: Enable: 0x201c[1:0]; Drive Strength: 0x2020[5:3]
```

- Adjustment and Range

Note: This register is multiplexed with TX PMOS current control.

The following io commands are for Lane0; for other Lanes, please modify the register offset. For example, the address for Lane1 is 0xfed8901c (0xfed8881c + 0x800).

Amplitude	Command [reg offset 0x081c, bit 7:5]
3'b111:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x78
3'b110:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x70
3'b101:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x68
3'b100:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x60
3'b011:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x58
3'b010:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x50
4'b001:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x48
4'b000:	io -4 0xfed8881c 0xe6 io -4 0xfed88820 0x40

2.4 DP TX SSC Control

2.4.1 PLL SSC Modulation Deviation

- Register Offset

```
// default=0x19
RBR: Adjustment: 0x03F0[5:0]
HBR: Adjustment: 0x03F4[5:0]
HBR2: Adjustment: 0x03F8[5:0]
HBR3: Adjustment: 0x03FC[5:0]
```

- Adjustment and Range

Level	Command [reg offset 0x03F0, bit5:0]
6'b000000:	io -4 0xfed883f0 0x00
6'b000001:	io -4 0xfed883f0 0x01
...	...
4'b111110:	io -4 0xfed883f0 0x3e
4'b111111:	io -4 0xfed883f0 0x3f

2.4.2 PLL SSC Modulation Frequency

- Register Offset

```
// default=0x0f
RBR: Adjustment: 0x0404[4:0]
HBR: Adjustment: 0x0408[4:0]
HBR2: Adjustment: 0x040c[4:0]
HBR3: Adjustment: 0x0410[6:2] (default=0x3c)
```

- Adjustment and Range

Level	Command [reg offset 0x0400, bit4:0]
5'b00000:	io -4 0xfed88400 0x00 (HBR3: io -4 0xfed88410 0x00)
5'b00001:	io -4 0xfed88400 0x01 (HBR3: io -4 0xfed88410 0x04)
...	...
5'b11110:	io -4 0xfed88400 0x1e (HBR3: io -4 0xfed88410 0x38)
5'b11111:	io -4 0xfed88400 0x1f (HBR3: io -4 0xfed88410 0x3c)

HBR3 requires a left shift by two bits.

2.5 DP TX AUX Amplitude Control

- Register Offset

```
// default=0x36
aux swing: 0x0024[6:3]
```

- Adjustment and Range

Level	Command [reg offset 0x0024, bit6:3] (bit[2:0] = 0x6(default))
4'b0000: 0 mVpp	io -4 0xfed88024 0x06
4'b0001:	io -4 0xfed88024 0x0e
4'b0010:	io -4 0xfed88024 0x16
4'b0011:	io -4 0xfed88024 0x1e
4'b0100:	io -4 0xfed88024 0x26
4'b0101:	io -4 0xfed88024 0x2e
4'b0110:	io -4 0xfed88024 0x36
4'b0111: 390 mVpp	io -4 0xfed88024 0x3e
4'b1000:	io -4 0xfed88024 0x46
4'b1001:	io -4 0xfed88024 0x4e
4'b1010:	io -4 0xfed88024 0x56
4'b1011:	io -4 0xfed88024 0x5e
4'b1100:	io -4 0xfed88024 0x66
4'b1101:	io -4 0xfed88024 0x6e
4'b1110:	io -4 0xfed88024 0x76
4'b1111: 880 mVpp	io -4 0xfed88024 0x7e

3. Code Parameter Configuration Adjustment

Kernel Driver code location:

```
drivers/phy/rockchip/phy-rockchip-usbdp.c
```

U-boot Driver code location:

```
drivers/phy/phy-rockchip-usbdp.c
```

The Register definitions for adjusting swing and pre-emphasis are as follows:

```
struct dp_tx_drv_ctrl {
	u32 trsv_reg0204;
	u32 trsv_reg0205;
	u32 trsv_reg0206;
	u32 trsv_reg0207;
};
```

The configuration Register offsets for different lanes are as follows:

	trsv_reg0204	**trsv_reg0205**	**trsv_reg0206**	**trsv_reg0206**
lane0	0x0810	0x0814	0x0818	0x081C
lane1	0x1010	0x1014	0x1018	0x101C
lane2	0x1810	0x1814	0x1818	0x181C
lane3	0x2010	0x2014	0x2018	0x201C

The DP link rate has four levels: RBR, HBR, HBR2, and HBR3. For these four levels, the following swing and pre-emphasis configuration arrays are defined:

When outputting via the DP standard port:

	RK3576 Configuration	**RK3588 Configuration**
RBR	rk3588_dp_tx_drv_ctrl_rbr_hbr_typec	rk3588_dp_tx_drv_ctrl_rbr_hbr
HBR	rk3588_dp_tx_drv_ctrl_rbr_hbr_typec	rk3588_dp_tx_drv_ctrl_rbr_hbr
HBR2	rk3588_dp_tx_drv_ctrl_hbr2	rk3588_dp_tx_drv_ctrl_hbr2
HBR3	rk3588_dp_tx_drv_ctrl_hbr3	rk3588_dp_tx_drv_ctrl_hbr3

When outputting via the Type-C interface:

	RK3576 Configuration	**RK3588 Configuration**
RBR	rk3588_dp_tx_drv_ctrl_rbr_hbr_typec	rk3588_dp_tx_drv_ctrl_rbr_hbr_typec
HBR	rk3588_dp_tx_drv_ctrl_rbr_hbr_typec	rk3588_dp_tx_drv_ctrl_rbr_hbr_typec
HBR2	rk3588_dp_tx_drv_ctrl_hbr2	rk3588_dp_tx_drv_ctrl_hbr2
HBR3	rk3588_dp_tx_drv_ctrl_hbr3	rk3588_dp_tx_drv_ctrl_hbr3

Each of the above arrays has a size of 4x4 and is used to store Register configuration parameters for different combinations of swing and pre-emphasis levels.

The SSC code configuration location is as follows:

```
static const struct reg_sequence rk3588_udphy_24m_refclk_cfg[] = {
	......
	{0x03f0, 0x0a}, {0x03f4, 0x07},
	{0x03f8, 0x07}, {0x03fc, 0x0c},
	{0x0404, 0x12}, {0x0408, 0x1a},
	{0x040c, 0x1a}, {0x0410, 0x3f},
	......
};
```

The AUX amplitude code configuration location is as follows:

```
static const struct reg_sequence rk3588_udphy_init_sequence[] = {
	......
	{0x0024, 0x6e},
};
```